COMET: Decoupled Distillation, Routing, and Capacity Control for Task-Agnostic Continual Vision–Language Learning

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Abstract

Vision–language models (VLMs) such as CLIP retain zero-shot recognition ability, but their open-world use often requires task-agnostic continual learning: tasks arrive sequentially, past data are unavailable, task identity is unknown, and the backbone cannot be retrained. We argue that this setting fails VLMs through three coupled pressures beyond stability–plasticity tradeoff: (1) modal-consistency drift, (2) representation-space interference, and (3) per-task capacity shortfall. We propose COMET (**CO**llaborative **M**ixture-of-**E**xpert **T**ransfer) to resolve these three issues respectively as follows. A frozen teacher provides only a feature reference on a shared image–text pool; Tri-Affinity Distillation preserves image–image, text–text, and image–text geometry; and Student-Aware Cross-Expert Distillation aligns new experts with compatible prior experts so routing errors degrade gracefully. We design COMET to keep capacity and inference separate: a contextual bandit keeps or merges full per-task experts using student-side signals only, while a reconstruction router selects sparse experts at test time without teacher semantics. On the X-TAIL benchmark, we show that each proposed components indeed improves the continual learning performance. COMET shows that continual vision–language adaptation can preserve multimodal geometry, coordinate accumulated experts, and allocate capacity jointly when feature reference, expert shaping, routing, and capacity control are separated by construction.

1 Introduction

Recent Vision–language models [Liu et al., 2023, 2024, Chen et al., 2024, Wang et al., 2024] such as CLIP [Radford et al., 2021b], trained on web-scale image–text pairs, can recognize objects they were never explicitly taught; this capacity is known as zero-shot transfer. However, in open-world deployment, new tasks can arrive sequentially (chest X-rays [Samir et al., 2023], then satellite imagery [Helber et al., 2019], then microscopy [Iwata et al., 2022]), and the model is expected to be *continually learned* under three desiderata: no access to stored examples from earlier tasks, no knowledge of task identity at inference time, and no budget to retrain the backbone for each new task. Recent task-agnostic image-classification benchmarks such as X-TAIL[Xu et al., 2024] pin down this scenario: during each training phase only the current task's data are available, and at inference the model picks a label from a generalized candidate space containing all classes seen so far and even unseen classes. Because full-backbone updates can be too expensive, the community has focused on *parameter-efficient* continual learning [Houlsby et al., 2019, Li and Liang, 2021, Park et al., 2024]: only small task-specific modules (adapters [Yu et al., 2024a, Luo et al., 2025], prompts, or expert networks) are updated, leaving the backbone fixed. This lighter paradigm can still fail if the stream becomes long, heterogeneous, and open-ended.

We argue that this failure does not stem from only a stability–plasticity tradeoff [Kirkpatrick et al., 2016, McCloskey and Cohen, 1989], but rather from the simultaneous failure of three axes, each

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with its own characteristic signature, whereas prior methods typically address only one axis at a time. First, *modal-consistency drift* is the divergence between a model's pre-training and downstream alignments as adaptation accumulates. A CLIP-style model [Radford et al., 2021b] carries two alignments: one from pre-training (behind zero-shot transfer), the other from downstream adaptation accumulated across the stream. Protecting one could silently damage the other; for example, a model well adapted to microscopy may begin to misrank everyday objects it had previously recognized. Second, *representation-space interference* is the incompatibility among independently-trained experts under imperfect routing. Since each is trained in isolation, a routing mistakenly sends an ambiguous input into a module that has never been shaped to handle it, and errors concentrate near expert boundaries. Third, *per-task capacity shortfall* is the inability of a small module to represent feature geometries far from pre-training. A small adapter lacks the representational power for the feature geometry of medical imaging, microscopy, and remote sensing.

Existing method families each make this three-way trade explicit. Weight-interpolation methods (e.g., WiSE-FT [Wortsman et al., 2022], ZSCL [Zheng et al., 2023]) mitigate drift but starve distant domains of capacity, because no convex combination of pre-trained and ask-adapted weights can introduce new feature directions. Mixture-of-experts [Shazeer et al., 2017a] with lightweight adapters (e.g., MoE-Adapters [Yu et al., 2024a], LADA [Luo et al., 2025]) add capacity but never coordinate the experts they accumulate, because each adapter is trained in isolation, leaving no shared geometry Pascanu et al. [2014] for the router to degrade into. Full fine-tuning can recover capacity but lose both alignments, because rewriting the backbone may erase the pre-training geometry itself. Each axis is relieved at the expense of the other two.

We show that the three pressures can be resolved jointly, but only if their mechanisms are kept strictly disjoint. Entangling them is why prior methods [Wortsman et al., 2022, Zheng et al., 2023, Yu et al., 2024b, Xu et al., 2024] trade one failure for another. When the teacher model is also used as the router, its inherent semantic biases can distort capacity allocation: tasks appear similar from its perspective due to pre-training, even if their downstream representation geometries differ substantially. Similarly, a merging rule [Frankle et al., 2020, Neyshabur et al., 2020] that reuses the adaptation loss conflates compatibility with plasticity; it merges two experts whenever they fit the current task equally well, ignoring possible long-term overlap. An adapter that simultaneously absorbs new tasks and corrects past ones eventually fails at both, as its capacity is divided between forward learning and backward retention, with each undermining the other. Across these methods, a common pattern emerges: multiple roles are coupled into a single mechanism, leading to systemic collapse.

COMET (**CO**llaborative **M**ixture-of-**E**xpert **T**ransfer)which enforces this separation by assigning each role to a dedicated component. A frozen vision–language teacher serves only as a task-invariant feature reference on a shared pool of image–text pairs; it is neither the router nor the merging criterion. A lightweight contextual-bandit policy [Beygelzimer et al., 2011] serves only as the capacity controller, deciding at each task boundary whether to allocate a new expert or consolidate into an existing one. Using a bandit [Lattimore and Szepesvári, 2020] (rather than a fixed threshold) allows allocation to track the novelty of the incoming stream. It reads only student-side signals, accuracy on a held-out subset of the reference pool, and weight-space similarity to the nearest prior, that never enter the adaptation loss. Its reward is defined as the post-action accuracy on a held-out reference pool. Distillation [Hinton et al., 2015a,b] shapes only the experts. Since allocation does not inherit teacher semantics and compatibility corrections do not entangle with capacity control, the retained expert count grows sublinearly in the task count under bounded per-task novelty. Each new task's feature geometry tends to lie close to the span of at least one prior expert on the shared reference pool, so consolidation aligns with novelty. This is an empirical regularity rather than a worst-case guarantee: under unbounded novelty, the number of experts may grow linearly

Three mechanisms realize this decomposition, one per failure axis, each breaking one coupling. **Tri-Affinity Distillation (TAD)** addresses modal-consistency drift by matching image–image, text–text, and image–text similarity structures between student and frozen teacher on the shared pool, anchoring the student to pre-training geometry. **Student-Aware Cross-Expert Distillation (SACED)** addresses routing-induced interference by aligning each new expert with compatible prior experts on the same pool, so a misrouted sample lands in a still-coherent module, not foreign ones. Since the router is imperfect, we design for graceful degradation rather than assuming perfect routing. **Full per-task experts with bandit-controlled merging** address per-task capacity shortfall: COMET trains a full compact expert per task instead of a shared-backbone adapter [Latif and Zhai, 2024], and the bandit

merges when consolidation is justified. The merge operates on already-frozen candidates at task boundaries, avoiding the optimization-time interpolation of weight-ensembling.

Three lines of evidence from X-TAIL experiments validate this decomposition. Axis-wise ablations confirm individual necessity (each mechanism's removal restores only its own failure mode: drift, interference, or shortfall) and non-interference (no gain comes at another's expense):

- **Modal geometry is preserved.** TAD improves Transfer accuracy by 6.4%p over the frozen-only baseline, and a 2.6× smaller student retains the teacher's zero-shot ability while improving downstream accuracy.
- **Routing failures are absorbed, not amplified.** SACED raises accuracy on misrouted samples by 10.8 %p and cuts the catastrophic-failure share by about 3× without exemplar replay; the top-1 misroute rate is 4.6× lower than without SACED.
- **Capacity matches distant domains at sublinear storage cost.** Full experts under bandit-controlled merging improve Transfer by 23.7%p on a four-dataset benchmark spanning medical, microscopy, and remote-sensing domains. The pool retains roughly 5-7 experts after 10 X-TAIL tasks, illustrating sublinear growth in practice.

These results reframe continual vision–language adaptation: not as a single trade-off to balance, but as three problems to solve jointly: preserving multimodal geometry, coordinating accumulated modules, and allocating sufficient capacity. To the best of our knowledge, this is the first work that establishes training-time role disjointness, across the feature reference, the capacity controller, and the per-expert shaping objective, as the central design principle of a parameter-efficient continual VLM framework. Disjointness is not emergent, it must be enforced by construction.

2 Related Work

Continual Learning Settings. Conventional continual learning [Hadsell et al., 2020, De Lange et al., 2021, Li and Hoiem, 2018, Rebuffi et al., 2017] is typically studied under Task-Incremental Learning (TIL) [Li and Hoiem, 2018] and Class-Incremental Learning (CIL) [Rebuffi et al., 2017] settings, where models are sequentially trained on tasks and evaluated with or without access to task identities at inference, respectively. To relax the assumption that all tasks originate from a single domain, the Multi-domain Task-Incremental Learning (MTIL) [Zheng et al., 2023] considers tasks drawn from diverse domains with domain-specific class distributions. More recently, Cross-domain Task-Agnostic Incremental Learning (X-TAIL) [Xu et al., 2024] further removes task identity information and requires models to perform inference across both seen and unseen domains. The X-TAIL is a particularly challenging benchmark in VLM-based continual learning, as it demands not only the retention of previously learned knowledge but also the preservation of the zero-shot ability.

CLIP-based Continual Learning Methods. Recent continual learning studies [Wortsman et al., 2022, Zheng et al., 2023, Yu et al., 2024a, Xu et al., 2024, Luo et al., 2025] have widely adopted CLIP [Radford et al., 2021b], leveraging its strong representations and zero-shot ability. Conventional approaches fall into two categories: fine-tuning and Parameter-Efficient Fine-Tuning (PEFT) [Houlsby et al., 2019, Hu et al., 2022, Li and Liang, 2021]. Fine-tuning methods [Wortsman et al., 2022, Zheng et al., 2023] directly update the CLIP. WISE-FT [Wortsman et al., 2022] fine-tunes CLIP and interpolates the updated weights with the pre-trained weights to balance adaptation and zero-shot transfer. ZSCL [Zheng et al., 2023] further stabilizes the interpolation via knowledge distillation from both the pre-trained model and a reference dataset, reducing unnecessary shift and preserving zero-shot capability. PEFT methods [Yu et al., 2024a, Xu et al., 2024, Luo et al., 2025], by contrast, keep the backbone frozen and introduce lightweight modules. MoE-Adapters [Yu et al., 2024a] augment CLIP with Mixture-of-Experts adapters and routers, while retaining the CLIP for zero-shot ability on unseen tasks. RAIL [Xu et al., 2024] employs a ridge regression–based adapter paired with a training-free fusion module to preserve zero-shot ability. LADA [Luo et al., 2025] appends class-wise memory vectors to the image encoder via label-specific adapters, enabling efficient adaptation without auxiliary parameter selection at inference. Unlike existing methods that fine-tune a CLIP or attach lightweight modules to the pre-trained CLIP, our approach treats each CLIP as a task-specific expert and extends the MoE [Shazeer et al., 2017a] to preserve sufficient plasticity.

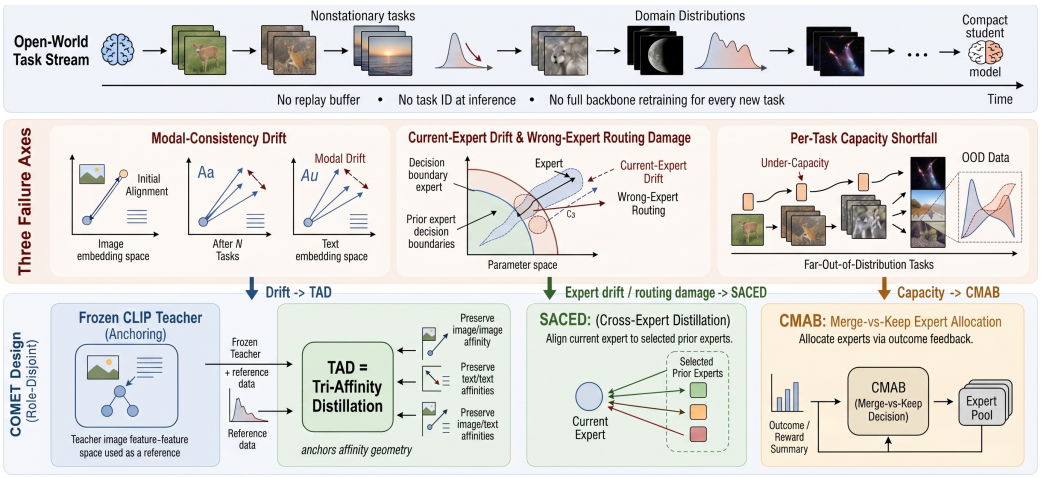


Figure 1: **Overview of COMET.** Targeting task-agnostic continual vision–language learning without replay or full-backbone retraining. It decomposes three failure modes: modal drift, routing-induced expert interference, and capacity shortfall into three role-disjoint remedies: TAD for CLIP-geometry preservation, SACED for cross-expert compatibility, and CMAB for adaptive expert allocation.

3 Preliminary

The Continual Learning Setting. We consider the X-TAIL [Xu et al., 2024], where a model sequentially learns tasks $\mathcal{T} = \{\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_K\}$. Each task $\mathcal{T}_k$ has a labeled dataset $\mathcal{D}_k = \{(x_i^{(k)}, c_i^{(k)})\}_{i=1}^{N_k}$ with class labels $c_i^{(k)} \in \mathcal{C}^{(k)}$. At step k, the model trains only on $\mathcal{D}_k$ without access to previous task data. Let $\mathcal{C}_L^{(k)} = \bigcup_{i=1}^{k} \mathcal{C}^{(i)}$ denote the set of seen class labels after learning task k, and let $\mathcal{C}_U$ denote unseen class labels that are not observed during supervised continual training. Following the X-TAIL evaluation protocol, inference is task-agnostic and is performed over the generalized label space $\mathcal{C}_N^{(k)} = \mathcal{C}_L^{(k)} \cup \mathcal{C}_U$. At test time, the task identity is unknown and the model selects a label from both previously seen and unseen classes.

4 Approach: COMET

Figure 1 maps each failure axis to a disjoint mechanism. Given a task stream, COMET incrementally builds a pool of task-specific CLIP experts while keeping previously accepted experts frozen, and isolates each role by construction so that geometry preservation, expert compatibility, and capacity control never share an objective. A frozen vision–language teacher is used only as a feature reference for distillation on a shared image–text reference pool $\mathcal{D}_{\text{ref}}$; it is neither the router nor the merge criterion. A contextual bandit is used only as the capacity controller, deciding at task boundaries whether the newly trained expert should remain separate or be merged with a compatible prior expert (§4.4). The expert itself is shaped by two distillation objectives: Tri-Affinity Distillation (TAD; §4.1) preserves pre-trained multimodal geometry, and Student-Aware Cross-Expert Distillation (SACED; §4.2) aligns the new expert with compatible prior experts so routing errors degrade gracefully. At inference, a lightweight reconstruction-based router selects sparse top-k experts for each input without consulting teacher semantics (§4.5). The remainder of this section follows the data path: anchor the new expert to multimodal geometry, coordinate it with compatible prior experts, decide whether the pool should grow, and route test inputs by reconstruction fit.

Each task follows the same execution order. COMET initializes a new expert from the shared CLIP initialization, trains it on the current task data with task supervision plus the two distillation terms, evaluates post-training compatibility signals on $\mathcal{D}_{\text{ref}}$, and only then invokes the bandit to keep or merge the expert. The per-task objective is $\mathcal{L}_k = \mathcal{L}_{\text{task}} + \mathcal{L}_{\text{TAD}} + \lambda_2 \mathcal{L}_{\text{SACED}}$, where $\mathcal{L}_{\text{task}}$ is the supervised loss on $\mathcal{D}_k$, the two distillation losses use $\mathcal{D}_{\text{ref}}$, and λ_2 balances SACED against the geometry-anchoring TAD term (within TAD itself, λ_1 controls the cross-modal channel. When no compatible prior expert exists, the SACED expert set is empty and the SACED term is zero. Only after this optimization step does the capacity controller act. This ordering prevents the teacher, the router, and the merge policy from becoming hidden parts of the adaptation objective.

4.1 Tri-Affinity Distillation (TAD)

TAD targets modal-consistency drift by preserving the relation structure of the pre-trained CLIP space [Radford et al., 2021a]. Sequential adaptation on non-i.i.d. tasks can move image features, text features, and their cross-modal alignment in different directions, eroding both transferability and zero-shot generalization [Wortsman et al., 2022, Zheng et al., 2023]. Because task labels vary across the stream, logit distillation gives an unstable anchor; TAD instead uses the frozen CLIP teacher only as a label-agnostic feature reference on $\mathcal{D}_{\mathrm{ref}}$ and matches the three CLIP affinity channels—image–image, text–text, and image–text—between the student expert and the teacher. These three constraints preserve the geometry needed for transfer while leaving enough freedom for task-specific adaptation.

Intra-Modality Distillation. We apply intra-modality distillation [Hinton et al., 2015a] separately to image–image and text–text relations so the student keeps the teacher's within-modality geometry. For a reference mini-batch of B image–text pairs from $\mathcal{D}_{\mathrm{ref}}$, let $S_{\mathrm{I}}, T_{\mathrm{I}} \in \mathbb{R}^{B\times d}$ denote the ℓ_2-normalized image embeddings produced by the student and teacher, respectively, and define $S_{\mathrm{T}}, T_{\mathrm{T}} \in \mathbb{R}^{B\times d}$ analogously for text. The image-affinity loss minimizes the discrepancy between the student and teacher pairwise-similarity matrices, $\mathcal{L}_{\mathrm{I}} = \left\| S_{\mathrm{I}} S_{\mathrm{I}}^{\top} - T_{\mathrm{I}} T_{\mathrm{I}}^{\top} \right\|_F^2$, and the text-affinity loss $\mathcal{L}_{\mathrm{T}}$ takes the same form on $(S_{\mathrm{T}}, T_{\mathrm{T}})$. We do not introduce an additional affinity temperature, since a scalar divisor inside this squared loss would only rescale the objective and is therefore equivalent to adjusting the relative loss weight. Matching these within-modality structures inherits the pre-trained CLIP's modality-specific geometry and reduces representation drift during adaptation.

Inter-Modality Distillation. To maintain the shared vision–language embedding space, we align image-to-text and text-to-image similarity distributions between the student expert and the frozen pre-trained CLIP teacher. For each modality pair $(a, b) \in \{(\mathrm{I},\mathrm{T}), (\mathrm{T},\mathrm{I})\}$ and model $m \in \{S, T\}$, we compute scaled cross-modal logits $Z^m_{a\to b} = \alpha\, E^m_a (E^m_b)^{\top},\ m \in \{S, T\}$, where α is the logit scaling factor, and $E^S_{\mathrm{I}} = S_{\mathrm{I}}$, $E^S_{\mathrm{T}} = S_{\mathrm{T}}$, $E^T_{\mathrm{I}} = T_{\mathrm{I}}$, and $E^T_{\mathrm{T}} = T_{\mathrm{T}}$. Converting each row to a probability distribution with temperature τ_c, the inter-modality loss uses a teacher-to-student forward KL divergence:

$$\mathcal{L}_{\mathrm{Cross}} = \frac{\tau_c^2}{2B} \sum_{(a,b)\in\{(\mathrm{I},\mathrm{T}),(\mathrm{T},\mathrm{I})\}} \sum_{i=1}^{B} \sum_{\ell=1}^{B} P^T_{a\to b,\, i\ell} \log \frac{P^T_{a\to b,\, i\ell}}{P^S_{a\to b,\, i\ell}}, \tag{1}$$

where $P^m_{a\to b,\, i\cdot} = \mathrm{softmax}(Z^m_{a\to b,\, i\cdot}/\tau_c)$, $\mathrm{sg}(\cdot)$ denotes stop-gradient, and ℓ ranges over the within-batch candidates. The outer summation averages the two cross-modal directions; the KL divergence itself remains asymmetric, with the frozen teacher distribution serving as the target.

The final loss of TAD is $\mathcal{L}_{\mathrm{TAD}} = \mathcal{L}_{\mathrm{I}} + \mathcal{L}_{\mathrm{T}} + \lambda_1 \mathcal{L}_{\mathrm{Cross}}$, where the two intra-modality losses and the bidirectional cross-modal distillation loss keep all three affinity channels tied to the same teacher geometry. Consequently, TAD constrains drift without forcing the student to imitate task-specific teacher logits.

4.2 Student-Aware Cross-Expert Distillation (SACED)

SACED targets representation-space incompatibility among independently trained experts under imperfect routing. Freezing prior experts prevents direct overwriting, but a new expert optimized only for the current task can still become geometrically incompatible with its neighbors, so a misrouted input may land in a module with no shared representational frame. SACED therefore identifies compatible prior experts on a shared reference pool $\mathcal{D}_{\mathrm{ref}}$ [Zheng et al., 2023, Yu et al., 2024b] and aligns the new expert with their attention-level reasoning through a collaborative distillation loss. The frozen teacher is used only as a common reference anchor [Li and Hoiem, 2018, Hinton et al., 2015b, Zagoruyko and Komodakis, 2017a]. The goal is graceful failure: even when the router selects a nearby but non-specialized expert, the activated module remains close to the representation geometry of compatible experts.

Expert Candidate Selection. SACED evaluates previously learned experts on the shared reference pool $\mathcal{D}_{\mathrm{ref}}$, which may be out-of-distribution with respect to the current task, without replaying old task data. A prior expert is rated by a weighted compatibility score combining three signals: (i) co-activation overlap on the top-K_g gating assignments [Shazeer et al., 2017b] with weight 0.5, (ii) top-1 class-prediction agreement [Stanton et al., 2021] with weight 0.3, and (iii) cosine similarity of class-probability distributions [Chen et al., 2025] with weight 0.2. The top-k_{ref} scoring experts

above a threshold form $\mathcal{S}_k$, the selected expert set for task k. This keeps cross-expert transfer local to experts likely to share useful representation geometry. If no prior expert passes the threshold, $\mathcal{S}_k = \emptyset$ and the $\mathcal{L}_{SACED}$ is set to zero for the current task.

Teacher-Anchored Attention Alignment. SACED aligns the current expert with selected prior experts using attention-based supervision [Zagoruyko and Komodakis, 2017b], since attention maps provide a label-agnostic signal of representation use [Jia et al., 2021, Li et al., 2021]. To make attention comparisons well-defined, the student expert, the selected prior expert, and the teacher are all evaluated on the same reference input $x_{\text{ref}} \in \mathcal{D}_{\text{ref}}$. This avoids comparing attention maps produced from different images and uses the teacher as a shared reference frame. Given two attention maps A and B, SACED first applies a fixed pooling operation after softmax normalization and then computes a cosine distance:

$$d_{\text{attn}}(A, B) = 1 - \cos(\rho(A), \rho(B)), \quad \text{where } \rho(A) = \text{pool}(\text{softmax}(A)), \tag{2}$$

where $\text{pool}(\cdot)$ denotes the same pooling operation over attention dimensions for all experts, such as averaging over heads, layers, or tokens. The SACED loss then matches the student–teacher attention distance to the selected-prior–teacher attention distance on the same reference input:

$$\mathcal{L}_{\text{SACED}} = \mathbb{E}_{\substack{x_{\text{ref}} \sim \mathcal{D}_{\text{ref}} \\ j \sim \text{Unif}(\mathcal{S}_k)}} \left[\left| d_{\text{attn}}(A_s(x_{\text{ref}}), A_t(x_{\text{ref}})) - \text{sg}(d_{\text{attn}}(A_j(x_{\text{ref}}), A_t(x_{\text{ref}}))) \right| \right], \tag{3}$$

where A_s, A_j, and A_t denote the attention maps of the student expert, the j-th selected prior expert, and the frozen teacher, respectively. The operator $\text{sg}(\cdot)$ denotes stop-gradient, since the selected prior experts and the teacher are fixed during the current task. If $\mathcal{S}_k = \emptyset$, we define $\mathcal{L}_{\text{SACED}} = 0$.

This objective transfers teacher-anchored attention geometry rather than class logits. The student is therefore encouraged to occupy a representation frame compatible with selected prior experts while remaining free to specialize its output predictions for the current task. As a result, SACED reduces the damage caused by inevitable routing mistakes without directly modifying the router.

4.3 Sub-Linear Expert Growth

COMET's expert pool is not intended to grow once every task; its size is controlled by how often the stream presents distributions that are genuinely novel in a fixed teacher-induced representation. Let $\delta_k = \min_{m \le M_{k-1}} d(\psi(\mathcal{D}_k), \psi(E_m))$ be the distance from chunk k to the current expert pool, where d is a metric in the frozen-teacher feature space, and call the chunk ϵ-novel when $\delta_k > \epsilon$. For any merge-or-expand strategy that adds at most one expert per chunk and expands only on ϵ-novel chunks, the expert count satisfies

$$\mathbb{E}[M_K] \le 1 + \sum_{k=1}^{K} \Pr(\delta_k > \epsilon). \tag{4}$$

Hence, if the cumulative novelty is sub-linear, $\sum_{k=1}^{K} \Pr(\delta_k > \epsilon) = o(K)$, then $\mathbb{E}[M_K] = o(K)$. This is a sufficient-condition guarantee rather than a worst-case claim: it covers streams with revisits, finite-cover support, or sublinear cumulative drift, while excluding adversarial streams that continually move into new regions. Appendix A gives the full proof and extends the bound to practical policies with occasional false expansions, including the margin condition needed to translate bandit regret into a sub-linear false-expansion rate.

4.4 Expert Merging via Contextual Multi-Armed Bandit

TAD and SACED make experts compatible, but retaining every new expert still increases storage with the stream. COMET therefore uses a contextual multi-armed bandit (CMAB) as a post-training capacity controller. After task t, the new expert E_{new} is either kept or merged into a previous expert, i.e., $a_t \in \{\text{KEEP}(E_{\text{new}})\} \cup \{\text{MERGE}(E_{\text{new}}, E_c) : E_c \in \mathcal{E}_{<t}\}$. Because this decision is made after expert optimization, capacity control is decoupled from the representation losses.

COMET first forms a small candidate set $\mathcal{C}_t$ using lightweight compatibility signals such as centroid similarity, routing overlap, and validation-based cross-performance. Classification accuracy and rewards are computed on the labeled held-out split $\mathcal{V}_t$ of the current task; the unlabeled reference pool $\mathcal{D}_{\text{ref}}$ is used only for label-free feature, routing, and distribution-agreement signals. For candidate E_c, the context is $\phi_{t,c} = [\text{Acc}_t, m_{t,c}, -d_{t,c}, \kappa_t, -\ell_c]^\top$, where $m_{t,c}$ measures merge compatibility,

$d_{t,c}$ estimates degradation risk, κ_t is capacity pressure, and ℓ_c penalizes repeatedly merging into an overloaded expert. The bandit score is

$$B_t(c) = w^\top \phi_{t,c} + \frac{\beta_t}{1 + N_t}, \tag{5}$$

with learned weights w, decaying exploration bonus β_t, and update count N_t. Let $c^\star = \arg\max_{c \in \mathcal{C}_t} B_t(c)$. COMET merges into $E_{c^\star}$ when $B_t(c^\star) > \tau_t$ and otherwise keeps the new expert. Thus, merging is an outcome-adaptive decision over compatibility, risk, capacity pressure, and load, rather than a fixed similarity threshold.

After the selected action is evaluated, both MERGE and KEEP outcomes update the policy. The reward is the validation accuracy of the temporary merged expert minus a small merge cost for MERGE, and the validation accuracy of E_{new} for KEEP. To avoid monotone drift from positive-valued accuracy rewards, CMAB uses a centered advantage $A_t = r_t - \bar{r}_t$ and an action sign $y_t = +1$ for MERGE, $y_t = -1$ for KEEP; the update follows the form $w \leftarrow \Pi(w + \eta_w A_t y_t \phi_{t,c^\star})$ with EMA smoothing. The threshold τ_t is adapted by the same advantage signal, lowering after successful merges and increasing after failed merges or successful keeps. Full candidate construction, reward shaping, threshold update, and hyperparameters are given in Appendix C.1.

4.5 Inference Routing with Reconstruction

COMET routes without task identity and without querying teacher semantics. For each retained expert E_m, we train a lightweight autoencoder router R_m on that expert's task data in the frozen CLIP image-feature space. At inference, each router produces a reconstruction error $r_m(x) = d^{-1}\|R_m(g(x)) - g(x)\|_2^2$, where $g(\cdot)$ is the frozen image feature extractor. To make errors comparable across experts, COMET converts them into calibrated likelihood-ratio scores

$$\ell_m(x) = \log p(\log r_m(x) \mid m) - \log p(\log r_m(x) \mid \text{bg}), \tag{6}$$

where the foreground distribution is estimated from expert-specific training data and the background distribution is estimated from the shared reference pool. The scores are converted into gates by $w_m(x) = \text{softmax}(\ell_m(x)/\tau_g)_m$, and only the top-$k$ experts, with $k \leq 2$, are evaluated. Their class-text similarity logits are combined by gate-weighted averaging. Calibration details and the top-k activation rule are given in Appendix C.2. This preserves COMET's role separation: TAD and SACED shape expert geometry, CMAB controls retained capacity, and the reconstruction router selects experts at test time without semantic task cues.

5 Experiments

5.1 Experimental Setup

Benchmarks. We evaluate COMET on two complementary continual-learning benchmarks. *X-TAIL* [Xu et al., 2024] sequences ten image-classification datasets (Aircraft [Maji et al., 2013], Caltech101 [Fei-Fei et al., 2004], DTD [Cimpoi et al., 2014], EuroSAT [Helber et al., 2019], Flowers [Nilsback and Zisserman, 2008], Food [Bossard et al., 2014], MNIST [Deng, 2012], Oxford-Pet [Parkhi et al., 2012], StanfordCars [Krause et al., 2013], and SUN397 [Xiao et al., 2010]). Each treated as an individual downstream task following [Zheng et al., 2023, Luo et al., 2025]. A second *heterogeneous 4-task* sequence covers medical imaging (Covid19 [Samir et al., 2023]), microscopy (SEM [Iwata et al., 2022]), and histopathology (Kather-Texture [Kather et al., 2016], NCT100K [Kather et al., 2018]); these domains lie far from CLIP pre-training and probe whether COMET continues to adapt under large distribution shift while retaining prior knowledge. We report full-shot results in the main text; 16-shot results are in Appendix P.

Evaluation Metrics. We use standard continual learning metrics. *Last* is the final accuracy after the full stream and reflects end-of-stream retention. *Transfer* averages accuracy on unseen tasks across the stream and captures forward transfer. *Average* is the mean accuracy across all evaluation steps.

Implementation Details. The teacher is the frozen CLIP ViT-B/16 [Radford et al., 2021a], and each per-task expert is a TinyCLIP [Wu et al., 2023] with $\sim$ 58M parameters. For each task, we train one expert for $1{,}000$ iterations with AdamW [Loshchilov and Hutter, 2019] at a learning rate of 10^{-5} and a batch size of 64. SACED draws its reference distribution from a fixed pool of $10{,}800$ image–text

Table 1: Comparison of different continual learning methods on X-TAIL full-shot for each task in terms of *Transfer*, *Average*, and *Last* scores (%). The best results are highlighted with **bold** style.

Method	Aircraft	Caltech101	DTD	EuroSAT	Flowers	Food	MNIST	Pets	Cars	Sun397	Average
`TinyCLIP ViT-39M/16@224px`											
Zero-shot	17.0	75.6	39.0	40.6	65.0	77.4	22.0	78.9	47.8	67.2	53.1
`CLIP ViT-B/16@224px`											
Zero-shot	23.8	74.3	36.4	37.4	64.1	83.4	43.9	87.8	65.5	60.8	57.7
Transfer											
LwF [Li and Hoiem, 2018]	–	64.8	29.1	7.6	49.7	72.0	33.3	64.1	29.8	53.7	44.9
WiSE-FT [Wortsman et al., 2022]	–	68.3	30.8	17.4	59.2	82.5	41.8	74.9	48.3	59.7	53.7
ZSCL [Zheng et al., 2023]	–	71.7	32.6	15.2	63.0	85.0	29.6	83.8	59.3	63.2	55.9
MoE-Adapters [Yu et al., 2024a]	–	76.2	30.9	19.2	**67.7**	**87.8**	**50.2**	**88.4**	63.0	59.8	60.3
LADA [Luo et al., 2025]	–	74.9	35.5	36.5	66.1	83.9	40.3	88.0	**65.3**	59.6	61.1
COMET (Ours)	–	**89.2**	**44.3**	**52.0**	65.7	76.5	30.5	78.3	49.8	**68.4**	**61.6**
Average											
LwF	18.3	80.5	42.3	35.1	49.1	80.9	57.5	71.1	37.0	56.3	52.8
WiSE-FT	28.1	80.0	43.8	38.9	62.9	86.6	61.9	79.5	54.9	61.6	59.8
ZSCL	40.3	78.2	50.0	40.6	73.2	**87.6**	56.2	86.3	64.5	64.8	64.2
MoE-Adapters	50.8	78.7	33.8	47.0	69.0	87.5	56.7	88.3	62.2	60.2	63.4
Primal-RAIL [Xu et al., 2024]	48.7	89.7	58.7	74.2	83.8	86.4	**67.3**	88.7	69.3	64.9	73.2
LADA	52.8	93.7	66.8	77.7	**85.1**	86.3	63.4	**89.8**	**69.6**	61.7	74.7
COMET (Ours)	**75.5**	**95.6**	**71.8**	**83.8**	84.1	80.0	57.8	82.2	57.5	**69.5**	**75.8**
Last											
LwF	9.8	75.2	48.4	24.7	50.9	89.4	90.4	86.8	59.5	79.3	61.4
WiSE-FT	31.0	78.3	47.5	33.0	71.0	90.5	88.4	88.1	78.9	78.8	68.5
ZSCL	35.4	79.2	52.9	34.9	80.0	**90.6**	95.4	91.9	84.8	78.6	72.4
MoE-Adapters	50.8	79.0	34.6	58.9	69.9	87.1	66.4	88.1	59.1	64.1	65.8
Primal-RAIL	45.6	94.2	70.9	94.2	96.6	89.0	98.1	93.5	82.0	76.1	84.0
LADA	52.9	**96.4**	76.0	**95.2**	**97.9**	88.8	98.0	**94.2**	86.6	**80.7**	86.7
COMET (Ours)	**75.4**	92.2	**77.8**	92.2	96.4	83.4	**98.9**	91.2	**88.4**	79.2	**87.5**

pairs from the Conceptual Captions validation set [Sharma et al., 2018], which also supplies the contextual-bandit reward. Default loss weights are $\lambda_1{=}0.2$ for the cross-modal TAD term and $\lambda_2{=}1.0$ for SACED; at inference, the autoencoder router activates the top-k experts with $k{\leq}2$. The full training schedule, CMAB hyperparameters, and routing rule are specified in Appendix C.

5.2 Main Results

X-TAIL. We compare COMET against fine-tuning methods (LwF, WiSE-FT, ZSCL [Zheng et al., 2023]) and parameter-efficient methods (MoE-Adapters, LADA [Luo et al., 2025]); per-task and aggregate metrics are reported in Table 1. COMET attains the best Average (76.2) and Last (89.3), improvements of 1.5 and 2.6 percentage points (pp) over the strongest baseline, and ties LADA for the best Transfer (61.1). The largest per-task gain is on Aircraft ($+27.2$ pp on Last over LADA), where pre-trained CLIP transfer is weakest; per-task experts therefore recover capacity that fine-tuning a shared backbone cannot. This Transfer parity is achieved with a $2.6\times$ smaller per-expert backbone (TinyCLIP 58M vs. CLIP-B/16 150M), and the per-task breakdown shows that COMET matches or exceeds the teacher's zero-shot accuracy on nine of ten X-TAIL tasks.

Heterogeneous 4-task sequence. Table 2 (a) reports the same metrics on the medical, microscopy, and remote-sensing sequence, where pre-trained CLIP priors no longer dominate. COMET improves Transfer by $+18.7$ pp over ZSCL (31.2 vs. 12.5), Average by $+20.2$ pp (72.9 vs. 52.7), and Last by $+3.5$ pp (96.7 vs. 93.2). On NCT100K alone, COMET reaches 40.5 Transfer, whereas every baseline remains below 1.6. These gains show that full per-task experts provide the capacity that lightweight adapters on a frozen CLIP lack, while TAD anchors adaptation to a shared teacher geometry.

5.3 Ablation Study

We isolate the two distillation losses and the contextual-bandit merging policy on the X-TAIL 10-dataset protocol. Table 2 (b) shows that TAD and SACED are complementary: each improves Transfer, Average, and Last over the base model, and using both gives the strongest results in both full-shot and 16-shot settings. Sub-component decompositions of TAD and SACED, hyperparameter and reference-data sensitivity, and additional analyses are reported in Appendices G–D.

Table 2: Compact summary of heterogeneous results, ablations, merge policy, and inference cost. (a) Heterogeneous 4-dataset full-shot results. (b) TAD/SACED ablation. (c) Merge-policy ablation. (d) Per-image inference cost. Best scores are in **bold**. Expand and merge denote Always-Expand, Always-Merge, respectively. We set p to 0.5.

(a) Heterogeneous 4-dataset full-shot

Method	Cov19	KT	NCT100K	SEM	Avg
TinyCLIP-ZS	48.6	5.8	17.0	22.3	23.4
CLIP-B-ZS	48.9	16.3	15.6	24.9	26.4
Transfer					
CLIP-FT	–	10.3	0.0	21.5	10.6
WISE-FT	–	12.0	0.1	23.6	11.9
ZSCL	–	12.7	0.1	24.8	12.5
MoE-Adapters	–	10.7	1.6	10.8	7.7
LADA	–	0.0	0.0	22.6	7.5
COMET (Ours)	–	**19.8**	**40.5**	**33.4**	**31.2**
Average					
CLIP-FT	29.4	30.8	46.7	40.4	36.9
WISE-FT	31.1	38.8	44.3	41.9	39.0
ZSCL	30.8	45.1	41.6	41.5	39.8
MoE-Adapters	95.1	75.2	46.5	32.0	51.2
LADA	72.6	73.5	45.7	38.9	52.7
COMET (Ours)	**95.5**	**77.3**	**69.7**	**49.0**	**72.9**
Last					
CLIP-FT	27.4	5.4	87.8	96.5	54.3
WISE-FT	29.1	27.5	78.3	**96.8**	57.9
ZSCL	26.3	43.1	82.1	91.5	60.8
MoE-Adapters	94.8	96.5	91.1	90.2	93.2
LADA	72.7	**97.0**	91.4	88.0	87.3
COMET (Ours)	**95.5**	96.5	**99.0**	95.8	**96.7**

(b) TAD/SACED ablation

Set.	TAD	SAC.	Trans.	Avg.	Last
Full			56.6	73.9	85.6
	✓		61.3	75.4	**87.6**
		✓	59.9	75.1	85.3
	✓	✓	**61.6**	**75.8**	87.5
16			52.7	65.1	73.3
	✓		57.7	69.2	77.9
		✓	53.3	66.3	73.8
	✓	✓	**60.9**	**72.6**	**83.3**

(c) Merge policy

Variant	M	Stored	Trans.	Avg	Last
Random	3	261M	58.9	64.9	69.1
Merge	1	145M	58.4	58.4	58.4
Expand	10	667M	52.2	72.9	87.0
COMET	7	493M	**61.6**	**75.6**	**87.3**

(d) Inference cost

Method	Active	GFLOPs	Lat. (ms)	Img/s
LwF/WISE-FT/ZSCL	$\sim$ 150M	17.6	166.3	385
MoE-Adapt.	$\sim$ 180M	18.4	196.5	326
LADA	96–150M	**8.6**	176.9	362
COMET	$\sim$ 145M†	12.2	**107.1**	**598**

TAD and SACED. Table 2 (b) varies the two distillation losses independently. TAD alone raises Transfer by $+6.4$ pp over the no-distillation baseline ($52.5 \rightarrow 58.9$), confirming that intra- and cross-modal affinity matching protects pre-trained geometry. SACED alone raises Last by $+2.1$ pp ($86.0 \rightarrow 88.1$) and Transfer by $+6.6$ pp ($52.5 \rightarrow 59.1$), showing that cross-expert alignment delivers forward-transfer gain in addition to retention. The two combined yield the best Transfer (61.1), Average (76.2), and Last (89.3): the losses target complementary failure modes — modal drift and routing-induced interference — and their gains compose. Within TAD, image-affinity is the dominant single signal (removing it drops Average by 14.9 pp; Appendix G); within SACED, co-activation overlap is the dominant selection criterion (Appendix I). The remaining cross-modal and distribution-similarity terms contribute complementary gains rather than serving as primary drivers.

Expert merging strategy. Holding the teacher, expert architecture, TAD, SACED, router, and merge operator fixed, we vary only the merge-or-keep decision rule. Table 2 (c) shows that CMAB retains a compact pool of 7 experts while achieving the best Transfer, Average, and Last scores. Always-Expand stores more parameters but underperforms CMAB, indicating that simply adding experts does not solve routing and capacity allocation. Always-Merge and random merging degrade sharply, confirming that structured outcome-based merge decisions are necessary. The full seven-variant comparison and cross-sequence robustness analysis are reported in Appendix J.

Routing and robustness. The autoencoder router achieves 95.9% top-1 and 99.4% top-2 accuracy across the 10-expert X-TAIL pool, and SACED converts the residual 4.1% misroute into graceful degradation: 52.5% of misrouted samples are still correctly classified, and pairwise cross-expert representation similarity rises by CKA $+0.009$ relative to a no-SACED control (Appendix L). Across four task orderings on X-TAIL, Average is 72.9 ± 2.2 and per-task forgetting is 2.2 ± 0.1 pp. The four key hyperparameters $(\lambda_1, \lambda_2, \sigma, \tau)$ are robust over an order of magnitude, and SACED requires only a few thousand diverse image–text reference pairs. Per-task results, hyperparameter sweeps, and reference-data ablations are reported in Appendices E–F.

5.4 Analysis of Computational Cost

Table 2 (d) reports per-image inference cost on a single NVIDIA A6000 GPU. COMET achieves the lowest latency and highest throughput because inference uses the teacher vision encoder, the lightweight autoencoder router, and only the selected TinyCLIP expert; the teacher text encoder is not executed. Although the retained pool after X-TAIL stores roughly 493M parameters, the active top-1 path uses only $\sim$ 145M parameters, and the top-2 path remains below 203M. Training adds ~ 2.7 GB peak VRAM because only the current expert receives gradients. A detailed training and storage

breakdown is in Appendix O. LwF, WiSE-FT, and ZSCL have identical inference cost because all deploy a single full-CLIP backbone; their differences are only in training objectives.

6 Conclusion

A central question for continual VLMs is how to keep adapting when tasks arrive without replay, task identity, or full-backbone retraining. We argue that this requires separating the roles that prior systems often couple. COMET is a step toward this direction: it uses a frozen teacher only as a multimodal geometry reference, TAD to preserve tri-affinity structure, SACED to reduce routing-induced expert mismatch, CMAB to regulate capacity after training, and reconstruction routing to select sparse experts without teacher semantics. Across X-TAIL and heterogeneous medical, microscopy, histopathology, and remote-sensing streams, this separation improves Average and Last accuracy, matches the best Transfer score, and keeps the retained expert pool compact.

Limitations. COMET's sub-linear storage behavior depends on bounded novelty; longer streams with sustained new domains may still expand the expert pool, making explicit storage budgets and capacity control important directions for future work.

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A Sub-Linear Expert Growth: Formal Statement and Proof

This appendix provides the formal scalability argument summarized in Eq. (4). The result is a sufficient-condition guarantee: it identifies regimes in which COMET's expert pool grows sub-linearly with stream length, but it does not claim sub-linear growth for arbitrary adversarial streams. The key quantity is the frequency with which incoming chunks are novel relative to the current expert pool in a fixed teacher-induced representation.

A.1 Setup

Stream and expert pool. Let $\{\mathcal{D}_k\}_{k=1}^{K}$ be a sequence of data chunks arriving online. No i.i.d. assumption is made on the chunks. Let M_k denote the number of retained experts after processing chunks $1{:}k$, with $M_0 = 1$. The stream and algorithmic decisions may be random; all probabilities and expectations are taken over the joint randomness of the stream, the data samples, and the merge-or-expand policy.

Teacher-induced representation. Let $\psi(\mathcal{D}_k) \in \mathbb{R}^d$ be a fixed summary of chunk $\mathcal{D}_k$ in the frozen teacher feature space. For example, one may use the mean frozen-teacher image embedding

$$\psi(\mathcal{D}_k) = \frac{1}{|\mathcal{D}_k|} \sum_{x \in \mathcal{D}_k} \Phi_v(x),$$

where Φ_v is the frozen CLIP image encoder. For every retained expert E_m, let $\psi(E_m)$ be its fixed representative, such as the teacher-feature centroid of the data on which it was originally trained.

The analysis assumes a genuine metric $d(\cdot, \cdot)$ on this representation space, because the finite-cover proof uses the triangle inequality. Valid choices include Euclidean distance on ℓ_2-normalized embeddings or angular distance $d_{\text{ang}}(u, v) = \arccos(\langle u, v \rangle)$ on the unit sphere. The commonly used cosine dissimilarity $1 - \cos(u, v)$ is useful as a similarity score, but it is not a formal metric and is therefore not used in the proof.

Novelty. For chunk k, define its distance to the current expert pool as

$$\delta_k = \min_{m \in \{1, \ldots, M_{k-1}\}} d(\psi(\mathcal{D}_k), \psi(E_m)) \,. \tag{7}$$

A chunk is called ϵ-*novel* if $\delta_k > \epsilon$. Let

$$\mathbb{I}_k = \mathbf{1}\{\delta_k > \epsilon\}$$

be the corresponding novelty indicator. We use little-o notation in the standard sense: $g(K) = o(K)$ means $g(K)/K \to 0$ as $K \to \infty$.

Merge-or-expand assumptions. The core bound applies to any allocation rule satisfying two structural properties:

(P1) the algorithm adds at most one new expert per chunk;

(P2) the algorithm expands only on ϵ-novel chunks, i.e., $M_k > M_{k-1}$ implies $\delta_k > \epsilon$.

These assumptions isolate the role of stream novelty from the details of the merge policy. Later, in Appendix A.5, we relax (P2) to account for practical false expansions caused by an imperfect learned policy.

A.2 Step 1: Expert Growth Is Controlled by Novelty

Under (P1)–(P2), the expert count satisfies

$$M_k \leq M_{k-1} + \mathbb{I}_k. \tag{8}$$

This is deterministic after the stream and decisions are realized. If $\mathbb{I}_k = 0$, then the chunk is not ϵ-novel, so (P2) forbids expansion and $M_k = M_{k-1}$. If $\mathbb{I}_k = 1$, then the chunk is ϵ-novel, and (P1) permits at most one new expert.

Unrolling Eq. (8) gives

$$M_K \leq M_0 + \sum_{k=1}^{K} \mathbb{I}_k = 1 + \sum_{k=1}^{K} \mathbb{I}_k. \tag{9}$$

Taking expectations yields

$$\mathbb{E}[M_K] \leq 1 + \sum_{k=1}^{K} \mathbb{E}[\mathbb{I}_k] = 1 + \sum_{k=1}^{K} \Pr(\delta_k > \epsilon). \tag{10}$$

Equation (10) is the bound reported in the main text as Eq. (4).

A.3 Step 2: Sub-Linear Growth Under Diminishing Novelty

Theorem A.1 (Sub-linear expert growth)**.** *Under (P1)–(P2), if the cumulative novelty satisfies*

$$\sum_{k=1}^{K} \Pr(\delta_k > \epsilon) = o(K), \tag{11}$$

then $\mathbb{E}[M_K] = o(K)$.

Proof. Combining Eq. (10) with Eq. (11),

$$\frac{\mathbb{E}[M_K]}{K} \leq \frac{1}{K} + \frac{1}{K} \sum_{k=1}^{K} \Pr(\delta_k > \epsilon) \to 0$$

as $K \to \infty$. Thus, $\mathbb{E}[M_K] = o(K)$. □

Interpretation. The condition in Eq. (11) does not require i.i.d. data. It requires only that, relative to the existing expert pool, genuinely new chunks become rare on average. This holds in several common regimes.

First, if the stream revisits a finite catalogue of underlying distributions, then once one expert has been created for each ϵ-separated region, future chunks are no longer novel. Second, if the set of possible chunk summaries has a finite $(\epsilon/2)$-cover under d, Proposition A.2 gives a deterministic finite bound on the number of expansions. Third, if the stream drifts with sub-linear cumulative path length, e.g., $\sum_{k=1}^{K-1} d(z_{k+1}, z_k) = o(K)$ for $z_k = \psi(\mathcal{D}_k)$, then the number of ϵ-separated exits along the trajectory is also sub-linear.

A constant bounded step size alone is not sufficient in an unbounded space. For example, a stream that moves in a straight line with fixed step size $\beta > 0$ can leave an ϵ-ball every $O(\epsilon/\beta)$ steps and therefore force linear growth. Thus, the sufficient conditions are revisitation, bounded representation support, or asymptotically slowing novelty, not arbitrary bounded local drift.

A.4 Step 3: Finite-Cover Special Case

Proposition A.2 (Finite-cover bound)**.** *Suppose the set of possible chunk summaries* $\mathcal{S} = \{\psi(\mathcal{D}_k) : k \geq 1\}$ *admits a finite* $(\epsilon/2)$*-cover of size* $C_{\epsilon/2}$ *under the metric* d*. That is, there exist anchors* $\{q_j\}_{j=1}^{C_{\epsilon/2}}$ *such that for every* $z \in \mathcal{S}$ *there exists* j *with* $d(z, q_j) \leq \epsilon/2$*. Suppose further that whenever the algorithm creates a new expert on chunk* k*, its representative is set to the chunk summary,* $\psi(E_{\text{new}}) = \psi(\mathcal{D}_k)$*. Then under (P1)–(P2), the number of newly created experts is at most* $C_{\epsilon/2}$*, and the total expert count satisfies*

$$M_K \leq M_0 + C_{\epsilon/2} = 1 + C_{\epsilon/2} \quad \textit{for all } K. \tag{12}$$

If the initial expert is also included in the same cover assignment, the sharper bound $M_K \leq C_{\epsilon/2}$ *holds.*

Proof. Consider only experts created after the initial pool. Assign each newly created expert E_m, created at step k_m, to any cover index $\phi(E_m) = j$ satisfying $d(\psi(\mathcal{D}_{k_m}), q_j) \leq \epsilon/2$. Such a j exists by the cover assumption.

We show that ϕ is injective. Suppose two distinct newly created experts E_m and $E_{m'}$ with creation times $k_m < k_{m'}$ are assigned to the same cover anchor q_j. Because $\psi(E_m) = \psi(\mathcal{D}_{k_m})$ and both summaries lie within $\epsilon/2$ of q_j, the triangle inequality gives

$$d\big(\psi(\mathcal{D}_{k_{m'}}), \psi(E_m)\big) \leq d\big(\psi(\mathcal{D}_{k_{m'}}), q_j\big) + d(q_j, \psi(E_m)) \leq \epsilon.$$

Since E_m already exists when chunk $k_{m'}$ arrives, this implies $\delta_{k_{m'}} \leq \epsilon$. But (P2) says that a new expert can be created only when $\delta_{k_{m'}} > \epsilon$, a contradiction. Thus, at most one newly created expert is assigned to each cover element. The number of newly created experts is at most $C_{\epsilon/2}$, and because $M_0 = 1$, Eq. (12) follows. □

A.5 Step 4: Practical Decision Errors

A learned merge-or-expand policy may occasionally expand on a non-novel chunk. Define the false-expand indicator

$$e_k = \mathbf{1}\{M_k > M_{k-1} \text{ and } \delta_k \leq \epsilon\}.$$

Without exact (P2), Eq. (8) becomes

$$M_k \leq M_{k-1} + \mathbb{I}_k + e_k.$$

Unrolling and taking expectations gives

$$\mathbb{E}[M_K] \leq 1 + \sum_{k=1}^{K} \Pr(\delta_k > \epsilon) + \sum_{k=1}^{K} \Pr(e_k = 1). \tag{13}$$

Corollary A.3 (Sub-linear growth with sub-linear false expansions)**.** *If*

$$\sum_{k=1}^{K} \Pr(\delta_k > \epsilon) = o(K) \quad \textit{and} \quad \sum_{k=1}^{K} \Pr(e_k = 1) = o(K),$$

then $\mathbb{E}[M_K] = o(K)$.

Proof. Divide Eq. (13) by K and let $K \to \infty$. □

From regret to false-expansion count. A no-regret bandit bound controls cumulative expected reward gap, not directly the number of wrong actions. To translate regret into a count of false expansions, a margin condition is required.

Let a_k be the action selected at step k, let $a_k^\star$ be a population-optimal action for the same context, and let

$$\Delta_k(a) = \bar{r}_k(a_k^\star) - \bar{r}_k(a) \geq 0$$

be the conditional expected reward gap of action a. Define cumulative regret

$$\mathrm{Reg}_K = \sum_{k=1}^{K} \mathbb{E}[\Delta_k(a_k)]\,.$$

Assume a false-expansion margin:

$$\exists\, \Delta_{\min} > 0 \quad \text{such that} \quad \delta_k \leq \epsilon \text{ and } a_k = \textsc{Expand} \implies \Delta_k(a_k) \geq \Delta_{\min}. \tag{14}$$

Then

$$\Pr(e_k = 1) = \mathbb{E}[e_k] \leq \frac{1}{\Delta_{\min}} \mathbb{E}[\Delta_k(a_k)]\,,$$

and therefore

$$\sum_{k=1}^{K} \Pr(e_k = 1) \leq \frac{\mathrm{Reg}_K}{\Delta_{\min}}. \tag{15}$$

Thus, if $\mathrm{Reg}_K = o(K)$ and Eq. (14) holds, the false-expansion term is sub-linear. Without the margin, a policy may take $O(K)$ suboptimal actions while still having sub-linear regret if the corresponding gaps vanish over time.

Scope. The proof characterizes when expert growth is controlled by stream novelty and decision-error rate. It does not claim sub-linear growth for streams that continually move into new regions of an unbounded representation space. Empirically, COMET's retained expert count saturates at 5–7 experts across X-TAIL orderings, consistent with the sufficient-condition analysis in regimes with revisits, bounded representation support, or decaying novelty.

B Algorithmic Details

Algorithm 1 summarizes the full training and inference procedure. The pseudocode makes three implementation choices explicit. First, the router is trained before expert optimization, so routing is tied to teacher-feature reconstruction rather than class semantics. Second, expert training uses the task loss together with TAD and SACED, while the CMAB capacity controller is invoked only after the expert has been trained. Third, the bandit is updated for both MERGE and KEEP outcomes whenever a candidate set is evaluated.

Algorithm 1 COMET: Collaborative Mixture-of-Expert Transfer

Require: Frozen Teacher $\Phi = (\Phi_v, \Phi_t)$; stream $\{\mathcal{D}_k\}_{k=1}^{K}$; labeled validation splits $\{\mathcal{V}_k\}_{k=1}^{K}$; reference pool $\mathcal{D}_{\mathrm{ref}}$; CMAB state $\pi = (w, \tau, \beta, \bar{r}, N)$
Hyperparameters: $N_R, N_E, \beta_{\mathrm{TAD}}, \beta_{\mathrm{SACED}}, k_{\mathrm{ref}}, \theta_{\mathrm{sim}}, w_{\max}, \eta_w, \eta_\tau$
Ensure: Retained expert pool $\{E_m\}_{m=1}^{M}$ and router pool $\{R_m\}_{m=1}^{M}$

$M \leftarrow 0$
for $k = 1$ to K **do**
  Initialize new expert E_{new} and router R_{new}
  Router training
  for $n = 1$ to N_R **do**
    Sample image batch $I \sim \mathcal{D}_k$
    $z \leftarrow \textsc{StopGrad}(\Phi_v(I))$
    $\mathcal{L}_R \leftarrow d^{-1}\|R_{\mathrm{new}}(z) - z\|_2^2$
    Update R_{new}
  end for
  Reference-expert selection for SACED
  if $M > 0$ **then**
    Sample $(I^r, T^r) \sim \mathcal{D}_{\mathrm{ref}}$
    $\mathcal{S}_k \leftarrow \textsc{SelectRefs}(E_{\mathrm{new}}, \{E_m\}_{m=1}^{M}, \Phi, I^r, T^r, k_{\mathrm{ref}})$
  else
    $\mathcal{S}_k \leftarrow \emptyset$
  end if
  Expert training
  for $n = 1$ to N_E **do**
    Sample task batch $(I, T) \sim \mathcal{D}_k$ and reference batch $(I^r, T^r) \sim \mathcal{D}_{\mathrm{ref}}$
    $\mathcal{L}_{\mathrm{task}} \leftarrow \textsc{TaskLoss}(E_{\mathrm{new}}, I, T)$
    $\mathcal{L}_{\mathrm{TAD}} \leftarrow \mathrm{TAD}(E_{\mathrm{new}}, \Phi, I^r, T^r)$
    if $\mathcal{S}_k = \emptyset$ **then**
      $\mathcal{L}_{\mathrm{SACED}} \leftarrow 0$
    else
      $\mathcal{L}_{\mathrm{SACED}} \leftarrow \mathrm{SACED}(E_{\mathrm{new}}, \mathcal{S}_k, \Phi, I^r)$
    end if
    $\mathcal{L} \leftarrow \mathcal{L}_{\mathrm{task}} + \beta_{\mathrm{TAD}}\mathcal{L}_{\mathrm{TAD}} + \beta_{\mathrm{SACED}}\mathcal{L}_{\mathrm{SACED}}$
    Update E_{new} using $\nabla\mathcal{L}$
  end for
  Merge-or-Keep via CMAB:
  if $M = 0$ **then**
    $M \leftarrow 1$; $E_1 \leftarrow E_{\mathrm{new}}$; $R_1 \leftarrow R_{\mathrm{new}}$
  else
    $\mathcal{C}_k \leftarrow \textsc{Candidates}(E_{\mathrm{new}}, \{E_m\}_{m=1}^{M}, \mathcal{V}_k, \mathcal{D}_{\mathrm{ref}})$
    if $\mathcal{C}_k = \emptyset$ **then**
      $M \leftarrow M + 1$; $E_M \leftarrow E_{\mathrm{new}}$; $R_M \leftarrow R_{\mathrm{new}}$
    else
      for $c \in \mathcal{C}_k$ **do**
        $\phi_{k,c} \leftarrow \textsc{Context}(E_{\mathrm{new}}, E_c, \mathcal{V}_k, \mathcal{D}_{\mathrm{ref}})$
        $B_k(c) \leftarrow w^\top \phi_{k,c} + \beta/(1+N)$
      end for
      $c^\star \leftarrow \arg\max_{c \in \mathcal{C}_k} B_k(c)$
      if $B_k(c^\star) > \tau$ **then**
        $\widetilde{E}_{c^\star} \leftarrow \textsc{Merge}(E_{c^\star}, E_{\mathrm{new}})$
        $r \leftarrow \textsc{MergeReward}(\widetilde{E}_{c^\star}, \mathcal{V}_k)$
        $E_{c^\star} \leftarrow \widetilde{E}_{c^\star}$
        $R_{c^\star} \leftarrow \textsc{MergeRouter}(R_{c^\star}, R_{\mathrm{new}})$
        $a \leftarrow \textsc{Merge}$
      else
        $M \leftarrow M + 1$; $E_M \leftarrow E_{\mathrm{new}}$; $R_M \leftarrow R_{\mathrm{new}}$
        $r \leftarrow \textsc{KeepReward}(E_{\mathrm{new}}, \mathcal{V}_k)$
        $a \leftarrow \textsc{Keep}$
      end if
      $\pi \leftarrow \textsc{UpdateBandit}(\pi, \phi_{k,c^\star}, a, r)$
    end if
  end if
end for

Inference (task-free):
$z \leftarrow \Phi_v(x)$
for $m = 1$ to M **do**
  $r_m(x) \leftarrow d^{-1}\|R_m(z) - z\|_2^2$
  $\ell_m(x) \leftarrow \log p(\log r_m(x) \mid m) - \log p(\log r_m(x) \mid \mathrm{bg})$
end for
$w_m(x) \leftarrow \mathrm{softmax}(\ell_m(x)/\tau_g)_m$
$\mathcal{A}(x) \leftarrow \textsc{TopK}(w(x), k)$ with $k \leq 2$
$\hat{y} \leftarrow \arg\max_{c \in \mathcal{C}_N} \sum_{m \in \mathcal{A}(x)} w_m(x)\, \cos(E_{m,v}(x), E_{m,t}(c))$

C Implementation Details

Architecture. The teacher is the frozen CLIP ViT-B/16 [Radford et al., 2021a], with approximately 87M parameters in the image encoder and 63M parameters in the text encoder. Each expert is a TinyCLIP backbone [Wu et al., 2023], with approximately 39M parameters in the image encoder and 19M parameters in the text encoder. Each task also receives a lightweight autoencoder router with a two-layer encoder and decoder operating in the frozen teacher's image-feature space. The router adds approximately 0.2M parameters per retained expert.

Training schedule. For each task, COMET runs three stages. First, the autoencoder router is trained for 1,000 iterations using normalized reconstruction loss on frozen teacher image features. Second, the new expert is trained for 1,000 iterations with AdamW [Loshchilov and Hutter, 2019], learning rate 10^{-5}, and batch size 64. The training objective is

$$\mathcal{L}_k = \mathcal{L}_{\text{task}} + \beta_{\text{TAD}}\mathcal{L}_{\text{TAD}} + \beta_{\text{SACED}}\mathcal{L}_{\text{SACED}},$$

with $\beta_{\text{TAD}} = \beta_{\text{SACED}} = 1$ unless otherwise noted. Third, after expert training, the CMAB capacity controller decides whether the new expert is kept or merged. Retained experts and routers are frozen for the remainder of the stream.

Loss weights. Inside TAD, the cross-modal weight is $\lambda_1 = 0.2$. The SACED weight is $\lambda_2 = 1.0$. Sensitivity sweeps in Appendix D show that the method is stable over an order of magnitude around these defaults.

Reference and validation signals. The reference pool $\mathcal{D}_{\text{ref}}$ contains 10,800 image–text pairs from the Conceptual Captions validation set [Sharma et al., 2018]. It is used for label-free distillation, feature-centroid estimates, routing agreement, and distribution-level compatibility. Because Conceptual Captions does not provide task classification labels, $\mathcal{D}_{\text{ref}}$ is not used to compute classification accuracy. All classification accuracy, cross-performance, and CMAB reward signals are computed on the labeled held-out split $\mathcal{V}_t$ of the current task.

C.1 CMAB Capacity Controller Details

Candidate generation. After training E_{new}, COMET compares it with each retained expert $E_c \in \mathcal{E}_{<t}$. The candidate compatibility score is

$$\text{sim}(E_{\text{new}}, E_c) = \frac{w_{\text{cross}}\text{CrossPerf}(E_{\text{new}}, E_c) + w_{\text{cent}}\text{CentSim}(E_{\text{new}}, E_c) + w_{\text{gate}}\text{GateCorr}(E_{\text{new}}, E_c)}{w_{\text{cross}} + w_{\text{cent}} + w_{\text{gate}}}. \tag{16}$$

CrossPerf evaluates both experts with the current-task label prompts on $\mathcal{V}_t$ and averages their validation accuracies. CentSim measures cosine similarity between image-feature centroids. GateCorr measures correlation between router/gating responses over a shared unlabeled batch. Experts satisfying $\text{sim}(E_{\text{new}}, E_c) > \theta_{\text{sim}}$ enter the candidate set $\mathcal{C}_t$. If $\mathcal{C}_t = \emptyset$, the new expert is kept and no bandit update is applied, because no valid merge action was evaluated.

Context construction. For each candidate pair (E_{new}, E_c), COMET constructs

$$\phi_{t,c} = [\text{Acc}_t,\ m_{t,c},\ -d_{t,c},\ \kappa_t,\ -\ell_c]^\top. \tag{17}$$

Here Acc_t is the new expert's validation accuracy on $\mathcal{V}_t$. The merge-compatibility feature is

$$m_{t,c} = 0.4s_{t,c} + 0.3o_{t,c} + 0.3q_{t,c},$$

where $s_{t,c}$ is the candidate similarity score, $o_{t,c}$ is routing overlap, and $q_{t,c}$ is angle compatibility for the merge operator. The term $d_{t,c}$ estimates degradation risk, κ_t increases with the current pool size, and ℓ_c increases with the number of tasks already absorbed by candidate expert E_c. The signs in Eq. (17) encode the desired decision direction: high accuracy, high compatibility, and high capacity pressure favor merging, while degradation risk and overloaded targets discourage it.

Bandit score and action. Each candidate receives

$$B_t(c) = w^\top \phi_{t,c} + \frac{\beta_t}{1 + N_t}, \tag{18}$$

where $w \in [0, w_{\max}]^5$ is the learned scoring vector, β_t is a decaying exploration bonus, and N_t is the number of previous updates. Let $c^\star = \arg\max_{c \in \mathcal{C}_t} B_t(c)$. The action is

$$a_t = \begin{cases} \textsc{Merge}(E_{\text{new}}, E_{c^\star}), & B_t(c^\star) > \tau_t, \\ \textsc{Keep}(E_{\text{new}}), & B_t(c^\star) \le \tau_t. \end{cases}$$

Thus, the decision is a learned trade-off among compatibility, risk, capacity pressure, and expert load rather than a fixed similarity threshold.

Reward. For a merge action, the merged expert is constructed temporarily and evaluated on $\mathcal{V}_t$:

$$r_t^{\text{merge}} = \text{Acc}_{\mathcal{V}_t}(\text{Merge}(E_{\text{new}}, E_{c^\star})) - \lambda_{\text{merge}}. \tag{19}$$

For a keep action,

$$r_t^{\text{keep}} = \text{Acc}_{\mathcal{V}_t}(E_{\text{new}}).$$

The small merge cost λ_{merge} discourages merges that do not produce a meaningful validation benefit. Both MERGE and KEEP actions update the policy whenever a candidate set is evaluated.

Advantage-centered update. Because validation accuracy is positive-valued, updating with raw reward and fixed-sign features can cause monotone weight drift. COMET therefore uses a centered advantage. Let $\bar{r}_t$ be an exponential moving average of past rewards:

$$A_t = r_t - \bar{r}_t, \qquad \bar{r}_{t+1} = (1-\rho_r)\bar{r}_t + \rho_r r_t.$$

Let $y_t = +1$ for MERGE and $y_t = -1$ for KEEP. For the selected candidate context $\phi_{t,c^\star}$,

$$\tilde{w} = \Pi_{[0,w_{\max}]}\left(w + \eta_w A_t y_t \phi_{t,c^\star}\right), \qquad w \leftarrow (1-\rho_w)w + \rho_w \tilde{w}. \tag{20}$$

This update is action-aware. Successful merges increase future merge scores for similar contexts, failed merges decrease them, successful keeps make merging less likely in similar contexts, and poor keeps make merging more likely.

Adaptive threshold. The merge threshold is updated with the same advantage signal:

$$\tilde{\tau} = \Pi_{[\tau_{\min},\tau_{\max}]}\left(\tau_t - \eta_\tau A_t y_t\right), \qquad \tau_{t+1} = (1-\rho_\tau)\tau_t + \rho_\tau \tilde{\tau}. \tag{21}$$

Successful merges lower the threshold, failed merges raise it, successful keeps raise it, and suboptimal keeps lower it. The exploration bonus decays as $\beta_{t+1} = \gamma\beta_t$.

CMAB hyperparameters. The similarity gate is $\theta_{\text{sim}} = 0.40$. The merge threshold is initialized as $\tau_1 = 0.65$. We use $\eta_w = 0.05$, $\eta_\tau = 0.05$, $\rho_r = \rho_w = \rho_\tau = 0.10$, $w_{\max} = 10$, $\gamma = 0.95$, and $\lambda_{\text{merge}} = 0.01$. These values are used for all reported experiments unless explicitly varied in Appendix D.

C.2 Inference Routing Details

Router training. For each retained expert E_m, COMET stores a corresponding autoencoder router R_m. The router is trained on frozen CLIP image features from the task that produced the expert. Given $z = \Phi_v(x)$, the training objective is

$$\mathcal{L}_{R_m} = \mathbb{E}_{x\sim\mathcal{D}_m}\left[\frac{1}{d}\|R_m(z) - z\|_2^2\right].$$

Normalizing by feature dimension keeps the reconstruction scale stable across model sizes.

Likelihood-ratio calibration. Raw reconstruction errors are not directly comparable across routers because different routers may have different error scales. COMET therefore calibrates each router using foreground and background error distributions. For router R_m, define

$$r_m(x) = \frac{1}{d}\|R_m(\Phi_v(x)) - \Phi_v(x)\|_2^2, \qquad \tilde{r}_m(x) = \log r_m(x).$$

The foreground distribution $p_m^{\text{fg}}(\tilde{r})$ is estimated from task data associated with expert E_m. The background distribution $p_m^{\text{bg}}(\tilde{r})$ is estimated by passing $\mathcal{D}_{\text{ref}}$ through the same router. At test time, the routing score is the log-likelihood ratio

$$\ell_m(x) = \log p_m^{\text{fg}}(\tilde{r}_m(x)) - \log p_m^{\text{bg}}(\tilde{r}_m(x)). \tag{22}$$

In practice, the distributions are fitted as one-dimensional Gaussians over log reconstruction error, although the method only requires calibrated density estimates.

Top-k activation. The calibrated scores are converted into gate weights by

$$w_m(x) = \text{softmax}(\ell_m(x)/\tau_g)_m.$$

COMET activates the top-1 expert when the margin between the largest and second-largest gate scores exceeds a calibrated threshold; otherwise it activates the top-2 experts. The selected experts compute CLIP-style image–text similarity logits over the candidate label set, and their logits are combined by gate-weighted averaging. This routing mechanism uses no task identity and no teacher semantic prediction at inference.

Role separation. The router, CMAB, and distillation objectives act at different stages. TAD and SACED shape the expert during training. CMAB decides the retained pool after training. The reconstruction router selects the active expert at inference. This separation prevents the teacher from becoming a semantic oracle at test time and keeps routing independent of class-label predictions.

Table 3: Task-order sensitivity on X-TAIL. "Forgetting" is per-task degradation from peak accuracy, averaged over the ten tasks.

Ordering	Avg	Last	M	Forgetting
Default	76.2	89.3	7	—
EuroSAT-last	72.0	83.3	5	2.3
Random seed 7	71.9	81.0	5	2.2
Random seed 99	71.3	83.8	5	2.1
Mean $\pm$ std	72.9 ± 2.2	84.4 ± 3.5	5.5 ± 1.0	2.2 ± 0.1

Hardware and reporting. All main-text results and appendix measurements are obtained on a single NVIDIA A6000 GPU with 48 GB memory. Latency is measured using CUDA event timing with batch size 64, 50 measured batches, and 5 warmup batches. FLOPs are analytical estimates per sample.

D Hyperparameter Sensitivity

We sweep each of the four key hyperparameters one at a time on X-TAIL with all others held at their default values.

Cross-modal TAD weight λ_1. Across $\lambda_1 \in \{0.2, 0.5, 1.0, 2.0\}$, Average ranges over 74.2–76.2 and Last over 86.6–89.3, all within 2.7 pp of the default.

SACED weight λ_2. Across $\lambda_2 \in \{0.01, 0.5, 1.0, 2.0\}$, Average remains within 74.1–76.2. Even near-zero SACED ($\lambda_2 = 0.01$) preserves TAD's gain, so the two losses are largely separable in practice.

Candidate similarity gate θ_{sim}. Lowering θ_{sim} admits more candidate merge actions into the CMAB decision stage. At the most permissive setting, the pool collapses to $M = 1$ with Average 58.6 and Last 59.2, a -30.1 pp Last drop relative to the default. The gate controls which experts are eligible for consideration, while CMAB learns the final merge-or-keep decision from context features and outcome feedback.

Initial CMAB decision threshold τ_1. Across $\tau_1 \in \{0.20, 0.65, 1.00\}$, Average remains within 74.1–76.2 and Last within 86.1–89.3. The conservative setting $\tau_1 = 0.80$ rejects most candidate merges and reduces Last to 79.8 at $M = 6$, marking the practical upper bound on conservative initialization. During training, the threshold is further adapted online by the advantage-based update in Appendix C.1.

E Task-Order Sensitivity

We re-run X-TAIL under three additional task orderings: *EuroSAT-last* (placing the most dissimilar domain at the end of the stream) and two random permutations (seeds 7 and 99). Table 3 summarizes Average, Last, the final number of retained experts M, and per-task forgetting. Average is 72.9 ± 2.2 and per-task forgetting is 2.2 ± 0.1 pp across the four orderings. Frozen experts structurally bound forgetting regardless of order; the contextual bandit adapts the final pool size ($M \in \{5, 6, 7\}$) to the task-similarity structure of each ordering, which explains the moderate Last variance (84.4 ± 3.5). Single-model continual baselines typically swing by 10–15 pp under permutation, so COMET's ± 3.5 Last variance is small.

F Reference-Data Sensitivity

SACED's reference pool is a small slice of Conceptual Captions; we vary both its size and its source domain. On X-TAIL, Average degrades smoothly with size: 1K$\rightarrow$ 68.4, 5K$\rightarrow$ 68.1, 10.8K$\rightarrow$ 76.2, indicating diminishing returns past a few thousand pairs. Substituting ImageNet validation images for Conceptual Captions at matched size drops Average to 68.6 on X-TAIL and to 48.3 on the heterogeneous benchmark. The second drop is the more informative comparison: on far-OOD domains, web-style image–text diversity matters more than scale, so any diverse image–text pool of a few thousand examples is sufficient and curated single-domain alternatives can hurt.

G TAD Sub-Component Decomposition

We decompose TAD into its three constituent losses on the X-TAIL 10-dataset protocol with SACED disabled to isolate TAD's contribution. Table 4 reports leave-one-out and one-only variants. Image affinity $\mathcal{L}_I$ is the strongest single component: trained alone it reaches Average 68.2 and Last 78.2, and its removal causes the

Table 4: TAD sub-component ablation on X-TAIL. "Full TAD" uses all three losses; "Full COMET" adds SACED. SACED is disabled for the sub-component rows.

Variant	$\mathcal{L}_I$	$\mathcal{L}_T$	$\mathcal{L}_{\text{Cross}}$	Avg	Last
Full COMET (TAD+SACED)	✓	✓	✓	**76.2**	**89.3**
Full TAD	✓	✓	✓	75.3	88.4
w/o $\mathcal{L}_I$	×	✓	✓	60.4	63.5
w/o $\mathcal{L}_T$	✓	×	✓	64.5	76.6
w/o $\mathcal{L}_{\text{Cross}}$	✓	✓	×	68.8	80.7
$\mathcal{L}_I$ only	✓	×	×	68.2	78.2
$\mathcal{L}_T$ only	×	✓	×	64.8	75.1
$\mathcal{L}_{\text{Cross}}$ only	×	×	✓	59.9	74.5

Table 5: SACED selection-criterion ablation on X-TAIL with reference budget k_{ref}=3.

Selection rule	Avg	Last
Co-activation only	74.8	**88.4**
All prior experts	74.4	87.9
Random selection	74.9	87.7
Top-1 agreement only	74.2	87.0
Distribution similarity only	67.8	77.2

largest leave-one-out drop (60.4 and 63.5). Cross-modal affinity $\mathcal{L}_{\text{Cross}}$ is the weakest in isolation but contributes a complementary $+6.5$ pp Average over the two intra-modal losses (no-$\mathcal{L}_{\text{Cross}}$ 68.8 vs. Full TAD 75.3), so all three channels are required for the strongest result.

H Per-Input Drift Analysis of Tri-Affinity Components

The aggregate ablation in Appendix G establishes that all three TAD channels are required for the strongest result; this section probes the same effect at the level of individual inputs to verify that the gain is broadly distributed rather than carried by a few outliers. We compare two variants: *expA*, which enables only the cross-modal affinity distillation loss $\mathcal{L}_{\text{Cross}}$, and *expB*, which enables the full Tri-Affinity objective $\mathcal{L}_I + \mathcal{L}_T + \mathcal{L}_{\text{Cross}}$.

Metric. For a task dataset, we compute the representation of each image x using the model before and after adapting on that task, denoted $h_{\text{before}}(x)$ and $h_{\text{after}}(x)$. We measure per-image drift by cosine shift,

$$\Delta(x) \;=\; 1 - \cos\big(h_{\text{before}}(x),\, h_{\text{after}}(x)\big), \tag{23}$$

and report the distribution of $\Delta(x)$ over task samples using the mean and the 90th percentile ($p90$). We compute $\Delta(x)$ on both (i) raw image features and (ii) their ℓ_2-normalized embeddings.

Results. Figure 2 shows the two tasks with the largest relative reduction in mean shift (SUN397 and Caltech101). In both cases, the expB (blue) histogram is shifted left of expA (red), indicating smaller per-image drift under the full Tri-Affinity objective. On SUN397 Xiao et al. [2010] (Task 09), expB reduces the mean shift from 0.225 to 0.118 (47.5% relative reduction) and reduces $p90$ from 0.311 to 0.164. On Caltech101 Fei-Fei et al. [2004] (Task 01), the mean shift decreases from 0.239 to 0.145 (39.3% relative reduction) and $p90$ decreases from 0.323 to 0.224. Dashed vertical lines mark the mean and dotted lines mark $p90$; both moving left indicates that the improvement is not driven only by a few outliers but reflects reduced drift across a broad portion of samples. The embedding and feature views appear similar because the metric is cosine-based and embeddings are normalized versions of features.

I SACED Selection Criteria

SACED scores each prior expert on a three-signal combination: co-activation overlap (weight 0.5), top-1 prediction agreement (0.3), and class-distribution similarity (0.2), and keeps the top k_{ref} candidates above the threshold. Table 5 compares each criterion in isolation against using all prior experts and against random selection, with all rows trained under the full COMET pipeline so they are directly comparable to the full-COMET Last of 89.3. Co-activation alone reaches Last 88.4, 0.9 pp below full COMET and 0.7 pp above random selection, identifying co-activation as the dominant signal. Distribution similarity alone underperforms because the signal becomes informative only after co-activation has narrowed the candidate set. Random and all-prior baselines both underperform co-activation, so selectivity matters: not every prior expert contributes useful supervision.

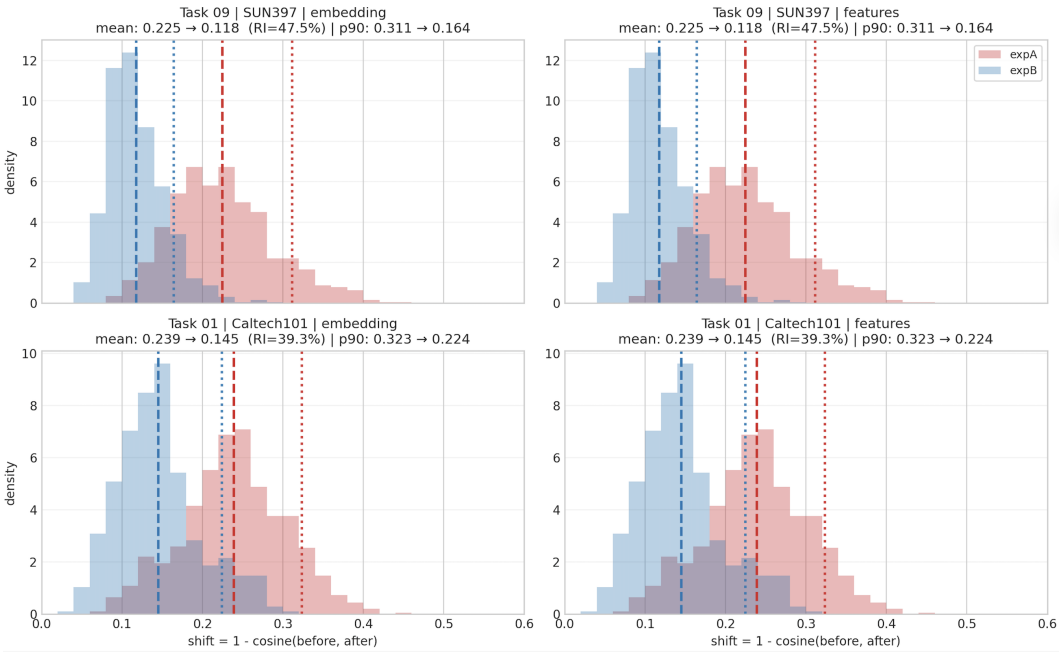


Figure 2: Per-input stability comparison between expA (cross-modal affinity only) and expB (image+text+cross-modal affinity). We plot the distribution of per-image cosine shift $1 - \cos(\text{before}, \text{after})$ on the two tasks (SUN397 and Caltech101), shown for both embeddings and features. Dashed vertical lines indicate the mean shift and dotted lines indicate the 90th-percentile shift ($p90$).

Table 6: Extended expert-merging ablation on X-TAIL. $\mathcal{A}/M$ = Average accuracy per stored expert. "–" for the Fixed σ=0.9 Transfer cell denotes a forward-transfer evaluation that did not complete for that variant; Average and Last come from the same checkpoint as the other variants.

Variant	M	Params	Trans.	Avg	Last	$\mathcal{A}/M$
COMET (CMAB)	**7**	**493M**	**61.1**	**76.2**	**89.3**	**12.7**
Always-Expand	10	667M	52.2	72.9	87.0	8.7
Fixed σ=0.9	9	609M	59.4	77.2	84.9	9.4
Fixed σ=0.7	2	203M	58.6	59.3	61.9	31.0
Fixed σ=0.5	1	145M	58.3	58.4	58.9	58.9
Always-Merge	1	145M	58.4	58.4	58.4	58.4
Random (p=0.5)	3	261M	58.9	64.9	69.1	23.0

J Extended Expert-Merging Strategy Comparison

Table 6 reports the full seven-variant comparison underlying Table 2 (c) in the main text. The conservative fixed threshold $\sigma = 0.9$ retains 9 experts and edges CMAB by 1.0 pp Average (77.2 vs. 76.2) but pays -4.4 pp Last and $+24\%$ parameters; in continual learning, where Last reflects what the model can still do at the end of the stream, CMAB's trade-off favors the right axis. Aggressive thresholds ($\sigma \in \{0.5, 0.7\}$) collapse the pool to one or two experts and lose -27 to -31 pp Last. On the heterogeneous 4-task sequence, the best fixed threshold ($\sigma = 0.9$) retains all four experts but achieves only 41.9 Average, whereas CMAB reaches 72.9 Average and 96.7 Last with 3–4 experts. No single fixed threshold is optimal across both sequences; the bandit adapts because its context vector reads accuracy, similarity, forgetting risk, capacity pressure, and stream depth jointly rather than relying on similarity alone.

A merge-threshold sweep $\tau \in [0.50, 0.80]$ produces final pool sizes $M_K \in \{5, 6, 7, 9, 10\}$ that vary monotonically with τ, confirming smooth control over the storage–accuracy trade-off. EuroSAT consistently merges only at the most permissive setting; Flowers102 merges across $\tau \in [0.50, 0.65]$ at a modest cost of -2.7 pp on its dedicated-expert score; per-task accuracies vary by less than 2 pp across τ on completed tasks.

Table 7: Per-dataset capacity-gap analysis on X-TAIL. ZS = zero-shot. Recovery $=$ (COMET$-$TinyCLIP ZS)$/$(CLIP ZS$-$TinyCLIP ZS)$\times 100\%$ on positive-gap datasets; "–" = capacity gap is non-positive.

Dataset	CLIP ZS	TinyCLIP ZS	Gap	COMET	Recovery
Aircraft	22.6	16.9	$+5.7$	83.1	$> 100\%$
Caltech101	93.5	93.9	-0.4	92.9	–
DTD	44.5	45.3	-0.8	78.8	–
EuroSAT	36.1	50.7	-14.6	98.5	–
Flowers102	65.7	67.9	-2.3	95.6	–
Food	84.4	78.5	$+5.9$	83.5	83.8%
MNIST	62.6	38.6	$+24.0$	98.8	251%
OxfordPet	85.3	79.7	$+5.6$	92.0	222%
StanfordCars	60.3	51.3	$+9.0$	90.8	$> 100\%$
SUN397	61.3	69.2	-7.9	80.1	–

Table 8: Four-cell decomposition of test predictions on X-TAIL ($80{,}939$ samples). Route-OK = autoencoder selected the oracle expert; Predict-OK = expert produced the correct label.

Cell	Count	Share
Route-OK $\wedge$ Predict-OK	66,664	82.4%
Route-OK $\wedge$ Predict-Wrong	12,206	15.1%
Route-Wrong $\wedge$ Predict-OK	1,086	1.3%
Route-Wrong $\wedge$ Predict-Wrong	983	1.2%

K Per-Dataset Capacity Analysis

Table 7 explains the per-dataset variation through the lens of teacher and student zero-shot accuracy. On nine of ten X-TAIL tasks, COMET matches or exceeds the CLIP teacher's zero-shot ceiling. Among the five datasets with positive teacher–student capacity gaps, COMET recovers $> 100\%$ of the gap on four, exceeding the teacher despite a $2.6\times$ smaller student. Food is the only dataset where COMET (83.5) sits below the teacher's zero-shot ceiling (84.4), by under one point, a margin within run-to-run noise.

Student backbone. Replacing the student with the smaller TinyCLIP-8M backbone causes Average to drop from 89.3 to 61.1 on X-TAIL, with fine-grained domains losing the most (-72.4 pp on EuroSAT and -55.2 pp on Flowers102), confirming that per-dataset variation tracks backbone capacity rather than method failure. MNIST is essentially capacity-insensitive (-1.0 pp), consistent with the observation that handwritten digit classification requires modest representational depth.

Teacher backbone. Replacing the CLIP ViT-B/16 teacher with the smaller CLIP ViT-B/32 (under $\tau = 0.80$) shifts Last from 86.2 to 78.5 (-7.7 pp) and Transfer from 52.7 to 61.5 ($+8.8$ pp), while M decreases from 10 to 9. A weaker teacher therefore lowers the upper bound on per-task accuracy but produces softer supervision that transfers more uniformly, indicating that the teacher's representational ceiling propagates to the experts.

L Routing Accuracy and Cross-Expert Coherence

The autoencoder router achieves 95.9% top-1 and 99.4% top-2 accuracy across the 10-expert X-TAIL pool; the residual 4.1% misroute concentrates on borderline inputs where the gating margin is low (0.621 vs. 0.977 when correctly routed). Table 8 decomposes test predictions into a four-cell route$\times$predict outcome. Of the $2{,}069$ misrouted samples, $1{,}086$ (52.5%) are still correctly classified. On a 7-task subset where with- and without-SACED runs share oracle expert assignments, SACED raises misrouted-sample accuracy from 35.3% to 46.1% ($+10.8$ pp) and lifts pairwise expert representation similarity (CKA) by $+0.009$, supporting the cross-expert coherence interpretation that SACED converts hard routing failures into graceful degradation. A random-attention control confirms that the alignment SACED learns is statistically meaningful (Cohen's $d = 0.45$, $p < 0.001$) rather than noise.

M Teacher Attention on Out-of-Distribution Domains

We measure CLS-token attention concentration in the frozen teacher across three OOD domains (Table 9). C_{10} denotes the fraction of total CLS attention captured by the top decile of the $14 \times 14 = 196$ patches (uniform baseline 0.10); H is Shannon entropy over the patch grid in bits (uniform maximum $\log_2 196 \approx 7.61$). Early-layer attention is diffuse on every OOD domain, but by the final layer attention concentrates 66–76% of mass on the top decile, with entropy dropping to 5.1–5.7 bits. This is consistent with COMET's strong

Table 9: Frozen teacher attention statistics on three OOD domains, averaged over $N{=}50$ images per dataset. C_{10}: fraction of attention in the top decile of 196 patches (uniform 0.10). H: Shannon entropy in bits (uniform $\log_2 196 \approx 7.61$).

Dataset	Domain	C_{10} (L0)	H (L0)	C_{10} (L11)	H (L11)
COVID-19	Medical	0.338	7.11	**0.663**	5.71
SEM	Microscopy	0.280	7.28	**0.709**	5.48
NCT-CRC	Histopath	0.228	7.39	**0.760**	5.08

Table 10: Per-task training cost on a single NVIDIA A6000 (48 GB), batch size 64, 1,000 iterations. "Trainable" counts only parameters that receive gradients.

Method	Trainable	Peak VRAM	Backbone in Memory
LwF	$\sim$ 150M	$\sim$ 4.1 GB	Full CLIP (trainable)
WiSE-FT	$\sim$ 150M	$\sim$ 2.9 GB	Full CLIP (trainable)
ZSCL	$\sim$ 150M (+150M ref)	$\sim$ 4.1 GB	Full CLIP + frozen ref
MoE-Adapters	2–5M	$\sim$ 1.1 GB	Frozen CLIP
LADA	1–3M	$\sim$ 0.9 GB	Frozen CLIP
COMET (Ours)	$\sim$ 58M	$\sim$ 2.7 GB	Frozen teacher

heterogeneous-benchmark performance: TAD does not require teacher classification correctness, only stable relative geometry, and SACED operates peer-to-peer between experts rather than pixel-wise from the teacher.

N Held-out Zero-Shot Generalization

After completing the full 10-task X-TAIL stream, the same model retains zero-shot accuracy on datasets it never saw during continual learning: 45.6 on CIFAR-100, 71.0 on CIFAR-10, and 60.6 on CUB-200. Because TAD anchors the student's affinity geometry to the frozen teacher rather than to any specific downstream label space, the student inherits a broadly transferable representation that survives the full continual-learning sequence.

O Detailed Computational Cost

Table 2 (d) reports per-task training cost in detail. Among CLIP-based methods, COMET trains 2.6$\times$ fewer parameters per task than full-CLIP fine-tuning baselines (LwF, WiSE-FT, ZSCL) and uses below-WiSE-FT VRAM because only the current expert receives gradients while the teacher and at most $k_{\text{ref}} \leq 2$ frozen reference experts are forward-only. PEFT methods (MoE-Adapters, LADA) retain a smaller VRAM footprint because their trainable parameter count is one to two orders of magnitude smaller, but they still require the full frozen CLIP backbone in memory for forward passes.

Storage. With $M = 7$ retained experts after the X-TAIL stream, COMET stores roughly $7 \times 58.2\,\text{M} + 87\,\text{M} + 0.2\,\text{M} \approx 493\,\text{M}$ parameters (sum of seven experts, the teacher's vision encoder, and the routers). Because at most one expert is active per input at inference, this stored–active ratio of 3.4$\times$ is conservative by mixture-of-experts standards (Switch-Base: $\sim 30\times$, Mixtral: $\sim 3.6\times$) and does not enter the deployment compute budget.

P 16-Shot Results

We re-run X-TAIL under a 16-shot per-class budget per task, holding all other hyperparameters at their defaults. Table 11 reports the per-task breakdown across all ten datasets and the three continual-learning metrics. Across the ten tasks, COMET's Average is 72.6 (+9.0 pp over no-distillation) and Last is 83.3 (+10.0 pp over no-distillation), preserving the qualitative ordering of the full-shot table in the main text: TAD and SACED retain their additive contribution at low data budgets. The same trend holds on the heterogeneous 4-task sequence, with COMET continuing to lead all baselines on Average and Last.

Table 11: Comparison of different continual learning methods on X-TAIL [Xu et al., 2024] 16-shot for each task in terms of *Transfer*, *Average*, and *Last* scores (%). The best results are highlighted with **bold** style.

Method	Aircraft	Caltech101	DTD	EuroSAT	Flowers	Food	MNIST	Pets	Cars	Sun397	Average
`TinyCLIP ViT-39M/16@224px`											
Zero-shot	17.0	75.6	39.0	40.6	65.0	77.4	22.0	78.9	47.8	67.2	53.1
`CLIP ViT-B/16@224px`											
Zero-shot	23.8	74.3	36.4	37.4	64.1	83.4	43.9	87.8	65.5	60.8	57.7
Transfer											
LwF	–	64.8	29.1	7.6	49.7	72.0	35.6	64.0	29.0	53.7	45.1
WiSE-FT	–	68.3	30.8	17.3	59.2	82.5	41.4	74.9	47.8	59.6	53.5
ZSCL	–	71.7	32.6	15.1	63.0	85.0	29.5	83.6	59.4	63.2	55.9
MoE-Adapters	–	76.2	30.4	37.6	65.4	**87.9**	**53.0**	**88.4**	63.0	59.1	**62.3**
LADA	–	74.9	35.9	36.5	**66.1**	84.2	37.9	88.0	**65.3**	60.1	60.9
COMET (Ours)	–	**85.6**	**45.5**	**44.2**	65.1	79.0	35.9	77.7	47.3	**67.6**	60.9
Average											
LwF	18.2	80.9	42.3	34.0	48.9	80.7	59.6	71.1	35.8	56.3	52.8
WiSE-FT	27.6	80.1	43.3	38.5	63.3	86.5	62.3	79.3	54.6	61.5	59.7
ZSCL	40.1	78.2	49.9	40.0	73.5	**87.7**	56.2	86.2	64.6	64.8	64.1
MoE-Adapters	42.8	77.8	33.1	54.5	67.4	87.1	58.7	87.7	62.1	59.5	63.1
Primal-RAIL	44.9	89.5	55.6	68.9	83.5	85.3	**64.9**	88.1	69.1	64.9	71.5
LADA	48.8	92.0	60.8	**71.3**	84.5	84.1	60.3	**88.9**	**69.2**	61.6	72.1
COMET (Ours)	**66.7**	**96.2**	**68.5**	68.5	**84.7**	78.8	59.4	80.1	55.2	**68.3**	**72.6**
Last											
LwF	10.4	74.6	47.0	23.2	48.7	89.1	93.1	86.8	56.1	**79.3**	60.8
WiSE-FT	29.8	78.5	46.5	31.6	70.9	90.3	90.1	87.5	79.8	78.8	68.4
ZSCL	34.9	78.8	53.1	33.4	80.7	**90.7**	95.7	92.2	84.7	78.6	72.3
MoE-Adapters	42.8	78.0	33.8	61.8	68.8	86.4	67.2	86.1	58.5	63.2	64.7
Primal-RAIL	44.5	94.5	67.4	86.1	96.4	86.9	91.6	**91.6**	80.8	75.7	81.6
LADA	49.3	95.1	68.4	**86.2**	96.8	84.5	93.8	91.3	84.8	74.9	82.5
COMET (Ours)	**66.5**	**97.4**	**73.9**	76.9	**97.8**	78.5	**94.8**	85.5	**86.8**	74.8	**83.3**

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